

EXPERIMENT NO. 1

Part A

AIM : Program on Class and Objects.

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Java (JDK 8 or above)
- IDE/Editor: IntelliJ IDEA / Eclipse / NetBeans / VS Code

Introduction:

Theory :

- **CLASS:**

A class is a group of objects which have common properties. It is a template or blueprint from which objects are created. It is a logical entity. It can't be physical.

A class is a blueprint from which individual objects are created.

A class in Java can contain:

- **Fields**
- **Methods**
- **Constructors**
- **Blocks**
- **Nested class and interface**

Syntax to declare a class:

```
class <class_name>
{
    field;
    method;
}
```

- **OBJECT:**

An entity that has state and behavior is known as an object e.g., chair, bike, marker, pen, table, car, etc. It can be physical or logical (tangible and intangible). The example of an intangible object is the banking system.

An object has three characteristics:

- **State:** represents the data (value) of an object.
- **Behavior:** represents the behavior (functionality) of an object such as deposit, withdraw, etc.

- **Identity:** An object identity is typically implemented via a unique ID. The value of the ID is not visible to the external user. However, it is used internally by the JVM to identify each object uniquely.

An object is an instance of a class. A class is a template or blueprint from which objects are created. So, an object is the instance(result) of a class.

Syntax to declare a object:

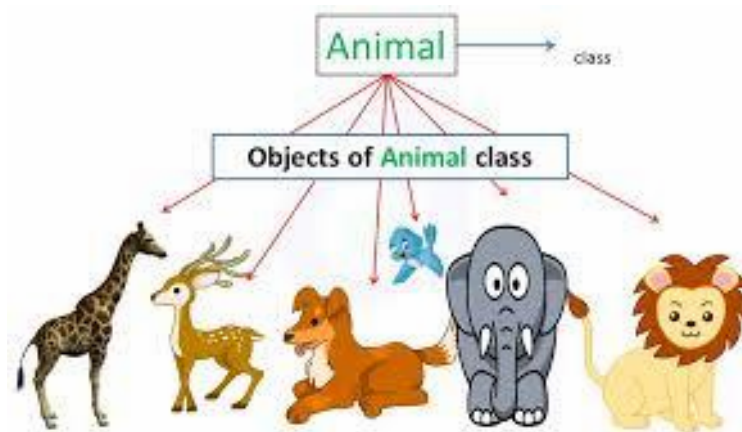
```
<Classname> object = new <Classname>();
```

There are 3 ways to initialize object in Java.

By reference variable

By method

By constructor



Code :

```
class Rectangle
{
    int length;
    int width;
    int area;
}
public class Shape
{
    public static void main(String args[])
    {
        Rectangle rect = new Rectangle();
        rect.length = 20;
        rect.width = 30;
        rect.area = rect.length*rect.width;
        System.out.println("Length: " + rect.length);
        System.out.println("Width: " + rect.width);
        System.out.println("Area: " + rect.area);
    }
}
```

```
}  
}
```

OUTPUT:

(Attach screenshots of your output screen)

CONCLUSION :

In the above experiment, we created a class Rectangle and an object. Here you may be surprised to see the main () method but don't worry it is just an entry point of the program by which JVM starts execution .Here, we used the main method to create object of Shape class and access its fields using Object.

Part B

Introduction:

Theory :

Aim: To implement default and parameterized constructors.

Constructor in java:

In Java, a constructor is a block of codes similar to the method. It is called when an instance of the class is created. At the time of **calling the constructor**, memory for the object is allocated in the memory. It is a special type of method which is used to initialize the object. Every time an object is created using the new () keyword, at least one constructor is called.

Rules for creating java constructor

There are basically two rules defined for the constructor.

1. Constructor name must be same as its class name
2. Constructor must have no explicit return type [Types of java constructors](#)

There are two types of constructors:

1. Default constructor (no-arg constructor)
2. Parameterized constructor

Java Default Constructor

A constructor is called "Default Constructor" when it doesn't have any parameter.

Java Parameterized Constructor

A constructor which has a specific number of parameters is called a parameterized constructor. Why use the parameterized constructor? The parameterized constructor is used to provide different values to distinct objects. However, you can provide the same values also.

OUTPUT:

```
J student.java J Experiment5.java X
Experiment5.java Experiment5
1 //0 Null
2 //Parameterized constructor
3
4 class Experiment5 {
5     int id;
6     String name;
7
8     int defid;
9     String defname;
10
11     // method to display the value of id and name
12
13     Experiment5(int i, String n) {
14         id = i;
15         name = n;
16     }
17
18     void display() {
19         System.out.println("Default constructor : " + defid + defname);
20
21         System.out.println("Parameterized constructor : " + id + name);
22     }
23
24     Run | Debug
25     public static void main(String[] args) {
26
27         // creating object
28
29         Experiment5 z1 = new Experiment5(i:9, n:" Jainisha");
30
31         z1.display();
32     }
33 }
```

CONCLUSION:

Learned and understood the use of default and parameterized constructors in Java.